

Problem-Solution Fit

Date	15/07/2026
Team ID	SWTID-2026-3095
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

1. CUSTOMER SEGMENT(S) [CS] Retail banks and NBFCs issuing credit cards; credit/loan officers who review applications; and individual applicants (salaried employees, commercial associates, pensioners, students) applying for a new credit card.	6. CUSTOMER LIMITATIONS [CL] Eg. Budget, Devices Smaller/regional banks have limited budget for expensive proprietary credit-scoring software; legacy core-banking IT systems make integrating a new ML model difficult; applicants in remote areas have limited access to digital/online banking channels.	5. AVAILABLE SOLUTIONS [AS] Pros & Cons Manual underwriting: Human judgement, but slow and inconsistent. Bureau credit score: Ready-made, but ignores the bank's own data. Plain Logistic Regression: Simple, but weaker here (~62% acc, 0.56 ROC-AUC).
2. PROBLEMS / PAINS [PR] + Its Frequency Every application (daily, high volume) needs manual review of financial and demographic details, which is slow and inconsistent across officers/branches; high-risk applicants with a history of 60+ days overdue are not consistently flagged before approval.	9. PROBLEM ROOT / CAUSE [RC] Applicant demographics and repayment history are stored separately with no unified, data-driven model linking the two, so decisions rely on rule-of-thumb judgement rather than measurable default-risk patterns.	7. BEHAVIOR [BE] + Its Intensity Officers manually cross-check each field before deciding; applicants repeatedly follow up on status. Intensity peaks during month-end and loan-campaign periods with very high application volumes.
3. TRIGGERS TO ACT [TR] A surge in incoming applications during festive/loan campaigns; rising default and delinquency rates prompting stricter screening; repeated applicant complaints about long waiting times.	10. SOLUTION [SL] An ML-powered Credit Card Approval Prediction System that merges applicant demographics with historical repayment records (by applicant ID), engineers features like age, years employed and income per person, then flags high-risk applicants as "Rejected" and the rest as "Approved". Logistic Regression, Decision Tree and Random Forest baselines were benchmarked; a tuned XGBoost classifier (RandomizedSearchCV + Stratified K-Fold CV) was chosen as the final model and saved with joblib for real-time deployment via a Flask web app.	8. CHANNELS OF BEHAVIOR [CH] ONLINE Web/mobile application forms, the bank's online portal, email and SMS status updates, the Flask-based prediction web app.
4. EMOTIONS [EM] Before / After Before: Applicants feel anxious waiting days for a decision; officers feel overwhelmed by paperwork and inconsistent judgement calls. After: Applicants feel confident getting an instant, transparent decision; officers feel confident relying on a standardized, data-driven evaluation.		